Interventions at the Cut

Pearl's do as a scoped guard in the rho calculus with graded where-clauses: an exact embedding of causal Bayesian networks, its σ-calculus generalisation, and the fragments that make it run

L. G. Meredith
F1R3FLY.io

F1R3FLY.io working note — November 2026
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Abstract

A program in the rho calculus with graded where-clauses, valued over the non-negative reals, already carries a causal model: the likelihood of a causal history is a product over firing sites of a locally normalised factor, each site's factor is its clause, each site's parents are the data its pattern and clause read, and the resolver that settles its race is the exogenous noise. This note makes the model explicit and adds the one thing Pearl's calculus has that the clause language lacked — an operator that intervenes.

The operator is a scoped decoration on parallel composition, $\mathsf{do}(\, y_1 \models \varphi_1 \,\&\, \cdots \,\&\, y_n \models \varphi_n \,)\ \mathsf{in}\ P$, read as: whenever a race for the binding of y_i arises inside P, resolve it among the candidates satisfying φ_i, with the site's own weights restricted to that set. It is placed at the cut because an aperture is a property of the cut and of neither agent flanking it, and because whether a particular cut will ever occur is undecidable even in the sequential fragment, so an intervention cannot be a surgery on a known node and must be a policy conditional on an event. The construct lowers into the where-clause by the normal-form theorem for weight maps, so it is a conservative extension, and it sits beside the cost membrane as a second index on a polymorphic cut.

With it, Pearl's causal Bayesian networks over finite domains embed exactly: a variable is a join on its parents' channels together with an ancilla race over its domain, the conditional probability table is the clause in disjunctive normal form, the term is staged so that each gate is its own contention set, and the truncated factorisation is the likelihood of a causal history. Intervening on the ancilla binding deletes the incoming edges on the nose: normalisation sends the one surviving weight to 1 for every value of the parents, and the positivity condition Pearl needs appears as a receipt that can starve. The three rules of the do-calculus are then theorems about the embedded fragment rather than axioms, and counterfactuals are replay from a recorded pre-state under the intervention. The construction is the one-qubit gadget of the quantum paper with real weights, and we say exactly what the resemblance amounts to. Restricting to a set rather than a point, or supplying a graded constraint, lands in Correa and Bareinboim's σ-calculus, whose induced graph we read off the free names of the constraint.

The naive interpretation of the construct is expensive in three independent ways — the constraint language can contain modalities and fixed points, the scope can be dynamic through reflection, and restriction changes the normalisation of contention sets whose enumeration is permanent-shaped — and whether a given do is vacuous is a reachability question. We propose fragments of the constraint language graded by evaluation cost, and type-based restrictions on the scope (staging, flatness, finite-domain channel types, namespace confinement) under which lowering is exact, checks are constant-time, and the semantics is Pearl's. We close by proposing the mechanism as the constructive content owed by the companion note on inhabiting choice: rung two of Pearl's ladder is a graded race under a do, rung three is its replay, and the vending-machine refusal that no trace can reveal is a receipt that starves under a one-line intervention.

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1 Introduction

1.1 Where the choice is

A rho program does not determine its own execution. Write

$$x!(Q_1) \mid \mathsf{for}(y \leftarrow x)P \mid x!(Q_2)$$

and the operational semantics permits two reductions without saying which happens or with what tendency. The graded where-clause programme [16, 17, 18] takes this silence as a parameter: the where clause of the for-comprehension is valued in a commutative semiring and evaluated once per candidate match, so that the programmer may write the resolution mechanism down rather than inherit it. Over the Booleans this is rholang; over $\mathbb{C}$ it is a syntax for arranging interference; over $\mathbb{R}_{\geq 0}$ it is stochastic execution, and the companion paper [18]

showed that the real instance is the one in which a learner lives, because it is the one with a likelihood to differentiate.

The present note observes that the real instance is also the one in which a *causal* model lives, in the precise sense of Pearl [1, 2], and that it has been living there since the clause was introduced. What was missing was not a model but an operator: the calculus could observe its races and weight them, but had no syntax for *forcing* one. Pearl's do is that syntax, and this note gives it a home.

1.2 The observation

The likelihood of a causal history in [18] is

$$\Pr_\theta(h) \;=\; \prod_{i=1}^{\ell} \frac{w(\theta,\sigma_i)}{\sum_{\tau\in\Gamma_i} w(\theta,\tau)}, \tag{1}$$

a product over firing sites of a factor normalised within the site's contention set Γ_i. Read with Pearl's dictionary this is a Bayesian-network factorisation. The variable is the site r and its value is the matching σ that fired. Its parents are the contents of Γ_r — what rests on the channels the pattern and the clause read. Its mechanism is the clause ψ_r. Its exogenous noise is the resolver's draw. And the thesis of the whole programme, that the resolution mechanism is a parameter supplied from outside the term, is Pearl's exogeneity: U is what the model does not explain.

Three consequences follow before any construction, and we state them here because they govern what follows.

- *The parent set is syntactic.* Pearl must posit the graph. Here the parents of a site are the names its pattern and the crisp subformulae of its clause read; the graph is written in the program. The one construct that can smuggle in an unlisted parent is reflection — $*y$ dropping received code — which is the same suspect the choice-formulae note [20] identified for subject reduction.

- *Observation and intervention are already priced differently.* Conditioning on "the race at r went x" keeps the histories in which it did and renormalises, which [16] showed is post-selection and charged at $1/\Pr(x)$ retries. Intervention replaces a column block of the transfer operator and renormalises nothing. The two differ exactly when histories producing x also influence the outcome by another route, which is a confounder; the paradigm confounder in the programme's own examples is a belief datum read by two sites.

- *The do-calculus is Paper II territory.* Pearl's rules presuppose a measure over histories. Over $\mathbb{C}$ a reachable outcome may carry amplitude zero, so there is no per-branch probability [18, Principle 3.5], and the counterpart on the complex side is the quantum-causal-models literature [11, 12], a different object. Everything below is over $\mathbb{R}_{\geq 0}$.

1.3 Contributions

1. **Interventions have one carrier** (§3). An intervention is a change of section over the candidate fibre of a site. The fibre at a deterministic site is a point, so deterministic communication is intervention-transparent (the dual of amplitude-transparency, [16, Prop. 8.5]); a context cannot force a race it does not own over $\mathbb{R}_{\geq 0}$ (the converse of the aperture principle), and can exactly in the idempotent algebras; and the observation/intervention gap — the whole content of Pearl's calculus — lives only on races.

2. **The construct** (§4). A scoped decoration on the cut, $\mathsf{do}(C)$ in P, with a membrane semantics and a lowering into the where-clause that agree on the reflection-free scope. It is a conservative extension by the normal-form theorem for weight maps, it composes by conjunction, and it is the operator that moves a formula from specification position to guard position in the sense of [19].

3. **The embedding** (§5). Causal Bayesian networks over finite domains are the $\mathbb{R}_{\geq 0}$ instance of the graded clause on staged terms; do on the ancilla binding is graph surgery on the nose (Theorem 5.7); the three rules are derived (Corollary 5.9); counterfactuals are replay (Theorem 5.11). Positivity appears as starvation.

4. **The Shor gadget** (Remark 5.6). The variable gadget is the one-qubit gadget of [17] with a stochastic matrix in place of a unitary one, and the resemblance is a theorem about the semiring rather than an analogy.

5. **The σ-calculus** (§6). Set-valued and graded constraints are Correa–Bareinboim σ-interventions [5]; the induced graph is read off the free names of the constraint; Pearl's atomic do is the singleton case.

6. **Complexity, and the fragments that tame it** (§7). The naive interpretation pays for modalities and fixed points in the constraint, for dynamic scope through reflection, and for permanent-shaped renormalisation of contention sets; vacuity of a do is a reachability question. We grade the constraint language into four fragments by evaluation cost and impose four type disciplines on the scope under which lowering is exact and the semantics is Pearl's (Proposition 7.4).

7. **Execution** (§8). The construct lowers in the elaboration pass the node already has, its membrane form is the seal tree's twin, an intervention is a COMM and is priced as one, and a counterfactual is the trace-replay the validators already perform.

8. **Inhabiting choice, discharged** (§9). The companion note [22] aligned Pearl's ladder with the linear/branching/causal spectrum by argument; this note supplies the constructive content, and the vending-machine benchmark becomes a one-line program.

1.4 What this note does not do

It does not treat continuous domains, which need the DDL's numeric types to carry an ancilla as a density. It does not prove the σ-calculus rules beyond the staged fragment, where the graph is dynamic and the right objects are the branching cells of [10]; that is stated as an obligation. It does not fix the token charge of a do against the ledger. And it does not treat the complex instance.

2 The calculus

The presentation is self-contained and follows [18]; a reader of that paper may skip to §3.

Definition 2.1 (Syntax)**.** Processes P and names x are given by

$$\begin{array}{rcl} P & ::= & 0 \mid x!(P) \mid \mathsf{for}(\vec{y} \leftarrow \vec{x}\ \mathsf{where}\ \psi)\, P \mid P \mid P \mid *x \mid \mathsf{new}\ x\ \mathsf{in}\ P \\ x & ::= & @P \end{array}$$

A receipt $\mathsf{for}(y_1 \leftarrow x_1 \,\&\, \cdots \,\&\, y_m \leftarrow x_m\ \mathsf{where}\ \psi)P$ is a *join*: it fires only when every pattern is matched. The clause ψ is drawn from the graded sort below and takes values in $\mathbb{V}$; its absence abbreviates $\lceil \top \rceil$. Receipts are linear by default; the persistent form $\Leftarrow$ survives its firing.

Definition 2.2 (Structural congruence)**.** $\equiv$ is the least congruence making $(|, 0)$ a commutative monoid and satisfying $*@P \equiv P$ and $@*x = x$, closed under α-equivalence and scope extrusion for `new`.

Definition 2.3 (Formula sorts)**.** The *crisp* sort retains the full propositional structure of the generated logic, including negation and fixed points:

$$\beta ::= \top \mid \mathrm{out}(n)\beta \mid \mathrm{in}(n)\beta \mid \beta \,|\, \beta \mid @\beta \mid \beta \wedge \beta \mid \neg\beta \mid \langle K\rangle\beta \mid \nu X.\beta.$$

The *graded* sort has no negation and takes values in $\mathbb{V}$:

$$\psi ::= \lceil\beta\rceil \mid v \mid \psi \otimes \psi \mid \psi \oplus \psi \mid \langle K\rangle_{\mathbb{V}}\psi \mid X \mid \mu X.\psi \mid \nu X.\psi,$$

with v ranging over constants of $\mathbb{V}$ and fixed-point variables positive and guarded. $\lceil - \rceil$ sends a crisp formula to 1 where it holds and to 0 where it does not.

Definition 2.4 (Candidates and contention)**.** Fix a term P and a receipt occurrence r with patterns $y_1, \ldots, y_m$ on channels $x_1, \ldots, x_m$. A *candidate* at r is an injective assignment σ of the patterns to messages resting on the corresponding channels; $\mathrm{Cand}(P, r)$ is the set of candidates. Two candidates *contend* if they share a message or share a linear receipt; a *contention set* is a connected component of the contention relation, and it is the unit at which weights are normalised.

Definition 2.5 (Weighted communication)**.** Write $\psi[\![\sigma]\!] \in \mathbb{V}$ for the value of the clause with the pattern variables bound as σ prescribes. If $\psi[\![\sigma]\!] \neq 0$ then $P \longrightarrow_{\psi[\![\sigma]\!]} P'$, where P' is the residue of consuming the assigned messages and r and substituting. A candidate of value 0 offers no transition.

Definition 2.6 (Completion over $\mathbb{R}_{\geq 0}$)**.** Throughout, $\mathbb{V} = (\mathbb{R}_{\geq 0}, +, \times, 0, 1)$. Resolution normalises within a contention set: the probability that a given maximal matching fires is its weight divided by the total over the set. A contention set is *complete* when that total is 1.

Definition 2.7 (Causal history)**.** A causal history h of length ℓ is a sequence of contention sets and maximal matchings $(\Gamma_1, \sigma_1), \ldots, (\Gamma_\ell, \sigma_\ell)$, quotiented by the interleaving of independent firings. Its likelihood is (1).

The quotient is load-bearing: without it m independent firings contribute the $m!$ linearisations of a partial order, measured as $2, 6, 24$ in the companion papers. With it the likelihood is a product over *events*, and an event structure's configurations are partial orders, hence acyclic. This is the first point at which the dictionary of §1 must be read with care: the *event-level* graph is always a DAG; the *variable-level* graph, which identifies events of a kind, may have cycles when the term recurses. Pearl applies verbatim at the event level. The variable level is where the fixed point of [18] lives, and where this note imposes types.

Proposition 2.8 (Support adequacy is exact over $\mathbb{R}_{\geq 0}$ [18, Prop. 3.3])**.** *With all clause constants strictly positive,* $\mathrm{supp}[\![P]\!] = \mathrm{Reach}(P)$*: every reachable outcome carries positive weight, so every branch of every contention set has a probability.*

3 Interventions have one carrier

Before adding syntax it is worth settling what an intervention can touch, since the answer constrains the syntax.

Definition 3.1 (Intervention, semantically)**.** An *intervention* at a site r of P is a replacement of the map $\mathrm{Cand}(P, r) \to \mathbb{V}$ denoted by ψ_r with another such map. It is *hard* when the replacement is the indicator of a single candidate, *set-valued* when it is the indicator of a subset, and *soft* otherwise.

Proposition 3.2 (Deterministic communication is intervention-transparent)**.** *At a site with exactly one candidate, every intervention is either the identity on the normalised dynamics or disables the site. In particular no intervention at such a site changes which outcome is reached when the site fires.*

Proof. The fibre is a singleton; a map on a singleton is determined by whether its value is zero. Normalisation within the contention set sends any nonzero value to 1. □

This is the dual of amplitude-transparency [16, Prop. 8.5]: a deterministic step neither carries weight nor admits a choice. To intervene at a deterministic site one must change the fibre — inject a datum or delete one — and the moment a datum is injected the site has two candidates and is a race. Deletion is the degenerate case, resolution toward the empty section.

Proposition 3.3 (No hard intervention by context over $\mathbb{R}_{\geq 0}$)**.** *Let r be a race in P. No rho context $C[-]$ makes $C[P]$ resolve r to a chosen candidate with probability 1 under the normalised $\mathbb{R}_{\geq 0}$ semantics, unless C consumes every competing candidate before r can fire — which is itself a race C wins with probability strictly less than 1. Over an idempotent resolution algebra (Viterbi, tropical) a context whose eater carries strictly dominating weight does force the outcome.*

Proof. The first claim is the aperture principle [16, Prin. 2.1] read in the forcing direction: a context can add contention and consume contenders, and consuming a contender is a candidate at a site in contention with r. Over $\mathbb{R}_{\geq 0}$ every candidate of positive weight has positive probability (Proposition 2.8), so r fires first with positive probability. Over an idempotent algebra resolution is argmax, which is attained by a definable section when the maximum is unique [20, Prop. 3.4(i)]. □

So the carrier of intervention is the candidate set at a receipt, a nontrivial intervention exists exactly where that set has or is given more than one element, and over $\mathbb{R}_{\geq 0}$ hard intervention requires access to the clause — white-box access, in the vocabulary of [21]. Two qualifications belong here. First, it is the *gap* between observation and intervention that lives only on races: one may replace $x!(1)$ by $x!(2)$ in a confluent program and change its output with no race anywhere, but that is running a different term, $\Pr(y \mid x)$ and $\Pr(y \mid \mathsf{do}(x))$ coincide trivially because there are no histories to reweight, and none of Pearl's rules has content there. Second, the clause controls the jump chain and not the holding times [18, Rem. 3.2], so an intervention on *when* a site fires is outside the clause and outside this note unless time is reified as data, at which point it is a race again.

Principle 3.4 (Interventions are metered because they are races)**.** A race is a COMM and a COMM costs one unit of phlo. Reading the trace is free; intervening is charged. This is not a policy choice: intervention has no carrier other than the event the ledger already prices.

4 The construct: a decorated cut

4.1 Why the cut

There is always a node in the syntax tree above a race, and it is a parallel composition: a race needs a receiver and at least two things it could meet. The graded programme already located the aperture there — an aperture is a property of the cut and not of either agent flanking it [16,

Rem. 2.3] — and the cost-accounting programme already made the cut polymorphic, indexing it by signatures and seals [24]. An intervention is a second index on the same operator.

The reason it cannot be anything more specific is a halting argument. Whether a particular cut will ever occur is undecidable, already in the sequential fragment, since rho is Turing complete and "this receipt fires" is a nontrivial reachability property. So an intervention cannot be a surgery on a node known in advance; it must be a *policy conditional on an event*: *if* a race for this binding arises, *then* resolve it thus. Vacuous if the race never arises. Pearl's nearest object is a dynamic treatment regime rather than an atomic do, and the formal home is the unfolding rather than the DAG.

4.2 Syntax

Definition 4.1 (The do membrane)**.** Extend Definition 2.1 with

$$P ::= \cdots \mid \mathsf{do}(\, C \,) \;\mathsf{in}\; P, \qquad C ::= y \models \varphi \mid y \models \psi \mid C \,\&\, C,$$

where y is a pattern variable, φ a crisp formula and ψ a graded formula, each with y among its free names. We write $\mathsf{do}(y = Q)$ for $\mathsf{do}(y \models \lceil y = @Q \rceil)$ and $\mathsf{do}(y \in S)$ for the disjunction over a finite set.

```
do( y = Q2 ) in { x!( Q1 ) | for( y <- x )P | x!( Q2 ) }

do( y1 |= phi1 & y2 |= phi2 ) in { P1 | P2 | P3 }
```

4.3 Semantics, twice

Definition 4.2 (Membrane semantics)**.** Let r be a receipt occurrence inside the scope of $\mathsf{do}(C)\ \mathsf{in}\ P$ binding a variable y constrained in C by φ (resp. ψ). At r the clause is read as

$$\psi_r \otimes \lceil \varphi \rceil \qquad (\text{resp. } \psi_r \otimes \psi),$$

with the constraint evaluated under the candidate's substitution, and normalisation within the contention set is taken *after* the restriction. Reduction is $\mathsf{do}(C)\ \mathsf{in}\ P \longrightarrow_w \mathsf{do}(C)\ \mathsf{in}\ P'$ whenever $P \longrightarrow_w P'$ under clauses so read. A receipt in scope binding an unconstrained variable is unaffected; a receipt arriving in scope by reflection is constrained like any other.

Definition 4.3 (Lowering)**.** On a scope P in which every receipt binding a constrained variable is syntactically present and its channel is a literal or new-bound name, define $\lfloor \mathsf{do}(C)\ \mathsf{in}\ P \rfloor$ to be P with each such receipt's clause replaced by $\psi_r \otimes \lceil \varphi \rceil$ (resp. $\psi_r \otimes \psi$), and the do erased.

Proposition 4.4 (Lowering is exact on flat scopes)**.** *If P contains no drop $*z$ at a position that can come to contain a receipt, then $\mathsf{do}(C)\ \mathsf{in}\ P$ and $\lfloor \mathsf{do}(C)\ \mathsf{in}\ P \rfloor$ have the same weighted transition system, hence the same graded denotation and the same likelihood on every causal history.*

Proof. Under the hypothesis every receipt that will ever be in scope is present in P up to substitution of names for pattern variables, which does not create or destroy binders. Definition 4.2 reads each such clause exactly as Definition 4.3 rewrites it, and the contention relation is unchanged because restriction removes candidates without adding any. □

The hypothesis is the *flat* fragment of [21], and the gap between the two semantics is exactly the dynamic scope that reflection introduces. We take the membrane as the definition and the lowering as the optimisation, for the same reason [20] stratifies where: received code is the one route by which an unlisted binder — and, in the causal reading, an unlisted parent — enters.

Remark 4.5 (Hygiene). $\mathsf{do}(y\ldots)$ in P refers to a name bound inside P, so y ceases to be α-convertible independently of the do. We read the construct as part of the binding scope: the do and every receipt it reaches are renamed together. The alternative — labelled receipts addressed by label — is better theory and worse ergonomics; the semantics is the same.

4.4 Restriction, override, and the normal form

Observation 4.6 (Restrict, not override)**.** Definition 4.2 *multiplies* the clause by the constraint. It does not replace it. For $\mathsf{for}(y \leftarrow x \ \mathsf{where}\ \lceil *y > 5 \rceil)P$ under $\mathsf{do}(y = 3)$, the restricted clause is 0 on every candidate and the site waits; an override would force a 3. Pearl's atomic do severs the mechanism and so is an override in spirit — but §5 shows that on the embedded models the two coincide, because the embedding puts the mechanism in the *weights* and the restriction is on an ancilla whose own gate is $\top$. Where they differ, restriction is the construct of this note and override is a σ-intervention (§6).

Proposition 4.7 (A compressed do is a weight map)**.** *A constraint $y \models \bigoplus_i v_i \otimes \lceil \chi_i \rceil$ with the χ_i pairwise exclusive denotes the weight map $(\chi_i, v_i)_i$ of the weighted-GSLT programme restricted to the fibre at y's site, and every weight map on that fibre is so denoted. In particular $\mathsf{do}(y = Q_1 \vee y = Q_2)$ is the weight map assigning 1 to two candidates and 0 to the rest.*

Proof. This is [16, Thm. 7.2] applied to the constraint rather than the clause. □

So several interventions compress into one expression for the reason weight maps have a normal form: a formula has one value. The do is a scoped weight map placed at the cut, and the normal-form theorem is what makes it a conservative extension.

Proposition 4.8 (Composition)**.** *$\mathsf{do}(C_1)$ in $\mathsf{do}(C_2)$ in P and $\mathsf{do}(C_1 \,\&\, C_2)$ in P have the same weighted transition system; $\&$ is commutative and associative; and $\mathsf{do}(y \models \top)$ in P has the transition system of P.*

Proof. Multiplication in $\mathbb{R}_{\geq 0}$ is commutative and associative with unit 1, and $\lceil \top \rceil = 1$. □

Remark 4.9 (Specification into guard). The omnibus [19, Part III] places one formula in four positions: specification, test, guard, hypothesis. A $\mathsf{do}(y \models \varphi)$ takes the ν-property "every binding of y satisfies φ" — a specification one would check — and installs it in guard position at every site in scope, including sites not yet visible. The do is the operator that turns a specification into a guard, and the Principle that the guard is where resolution becomes legible is why that is the right place for an intervention to live.

5 The embedding of Pearl's calculus

5.1 Causal Bayesian networks

Definition 5.1 (Causal Bayesian network)**.** A *causal Bayesian network* $M = (G, \{P_i\})$ over variables $V_1, \ldots, V_n$ with finite domains $D_1, \ldots, D_n$ consists of a DAG G on the V_i and, for each i, a conditional probability table $P_i(v \mid pa)$ for $v \in D_i$ and $pa \in \prod_{j \in \mathrm{PA}_i} D_j$. It induces $\Pr_M(v_1, \ldots, v_n) = \prod_i P_i(v_i \mid pa_i)$. M is *positive* when every table entry is strictly positive. For $X \subseteq V$ and an assignment x, the *submodel* M_x replaces P_i by the point mass at x_i for each $V_i \in X$ and deletes the incoming edges of X, and $\Pr_M(y \mid \mathsf{do}(x)) := \Pr_{M_x}(y)$.

A structural causal model refines this by writing each $V_i = f_i(\mathrm{PA}_i, U_i)$ with U_i independent exogenous variables; a CBN is an SCM with the U_i marginalised. The embedding below is of the CBN; §5.4 recovers the U_i as the record of the resolver's draws.

5.2 The variable gadget

Definition 5.2 (Fan-out)**.** For a variable V_i with children $V_{j_1}, \ldots, V_{j_k}$, allocate fresh channels $v_i^{(j_1)}, \ldots, v_i^{(j_k)}$, one per child, and a readout channel v_i°. Emitting V_i's value d is the parallel composition $v_i^{(j_1)}!(d) \mid \cdots \mid v_i^{(j_k)}!(d) \mid v_i^{\circ}!(d)$.

Definition 5.3 (The gadget)**.** For each V_i with parents $V_{p_1}, \ldots, V_{p_m}$ and domain $D_i = \{d_1, \ldots, d_s\}$, allocate a fresh ancilla channel w_i and write

```
new wi in {
    wi!(d1) | ... | wi!(ds)
  | for( p1 <- vp1_i & ... & pm <- vpm_i
       & u1 <- wi & ... & us <- wi
         where  [ *p1 = pa1 and ... and *pm = pam and *u1 = v ] * P_i(v | pa)
              + ...   // one summand per (pa, v) in the table
       )
    { emit_i( *u1 ) }
}
```

where `vpj_i` is the copy $v_{p_j}^{(i)}$ of parent j allocated to this gadget, $emit_i$ is the fan-out of Definition 5.2, the crisp bracket is the indicator embedding, and the sum has one summand for each $pa \in \prod_j D_{p_j}$ and $v \in D_i$. A root variable has $m = 0$. The *model term* $[\![M]\!]$ is the parallel composition of the gadgets under one `new` binding all the channels.

The join consumes all s ancilla messages, so there is no residual on w_i and no which-path record; the candidates are the $s!$ bijections from $u_1, \ldots, u_s$ to D_i, and the continuation mentions only u_1, so the $s!$ candidates fall into s outcome classes of $(s-1)!$ each, one per value of u_1. The clause is constant on each class. Under normalisation within the contention set the $(s-1)!$ multiplicity cancels, and the probability that the gadget emits v given parent data pa is $P_i(v \mid pa)$.

Lemma 5.4 (Staging)**.** *In $[\![M]\!]$ every receipt is its own contention set, each gadget fires exactly once, and the causal order of every history is a linear extension of G.*

Proof. Every channel has exactly one receiver (fan-out gives each child its own copy) and every ancilla channel is private to its gadget, so no two receipts share a message or a receipt; contention sets are singletons. A gadget is enabled exactly when all its parent copies have been emitted, which is [16, Prop. 8.3]: the circuit's DAG is the term's causal order. □

Theorem 5.5 (Adequacy of the embedding)**.** *Causal histories of $[\![M]\!]$ reaching a normal form are in bijection with assignments $(v_1, \ldots, v_n) \in \prod_i D_i$, read off the readout channels, and the likelihood* (1) *of the history for $(v_1, \ldots, v_n)$ is $\Pr_M(v_1, \ldots, v_n)$.*

Proof. By Lemma 5.4 a history is determined, up to the interleaving the quotient removes, by the outcome class chosen at each gadget, i.e. by the vector of values; conversely every vector is realised since every ancilla value is present. The factor contributed by gadget i is its restricted-and-normalised clause value, $P_i(v_i \mid pa_i)$ by the computation following Definition 5.3, and (1) multiplies the factors. □

Theorem 5.5 says the truncated-factorisation *form* is not imposed on the semantics; it *is* the semantics, and the staging that makes each gadget its own contention set is what makes the event structure confusion-free in the sense of [9], which is the condition under which local probabilistic choice composes to a global measure at all.

Remark 5.6 (The Shor gadget). Definition 5.3 is the one-qubit gadget of [16, Lemma 8.1] and [17, §10] — ancilla channel carrying the domain, a join that consumes everything, a continuation that re-emits without mentioning what was consumed, and a clause in disjunctive normal form indexed by (input, output) — with three changes: the domain is D_i rather than $\{0,1\}$, the inputs are several parents rather than one wire, and the matrix the clause realises is *stochastic* rather than unitary. The resemblance is exact and is a statement about the semiring. The quantum gadget realises every 2×2 matrix over $\mathbb{C}$ because the clause may mention both the input datum and the ancilla assignment; the present gadget realises every stochastic matrix over $\mathbb{R}_{\geq 0}$ for the same reason, and the restriction to stochastic rather than arbitrary non-negative matrices is supplied by normalisation within the contention set rather than by the clause. The two erasures of [16, Rem. 8.2] are both present: the join's totality erases which ancilla message was which (recombination within a superposition, inert here since there is no phase to recombine), and the continuation's silence about the parents erases the input basis (which is what makes the gadget a linear map into a fixed codomain, i.e. a conditional probability table rather than a family of unrelated tables). What differs is what the two calculi then do with the matrix: over $\mathbb{C}$ one arranges for columns to interfere and reads a detector that never fires; over $\mathbb{R}_{\geq 0}$ one reads the trace. Modular exponentiation in Shor is ordinary rholang running inside every branch at no cost [16, Cor. 8.6]; here the analogous statement is Proposition 3.2, that deterministic subcomputations of a causal model neither carry probability nor admit intervention.

5.3 Surgery

Theorem 5.7 (Graph surgery is restriction of the ancilla race)**.** *Let M be positive and $V_i \in V$, $v \in D_i$. Then*

$$[\![\, \mathsf{do}(u_1 = v) \text{ in } [\![M]\!] \,]\!] \;=\; [\![M_{V_i=v}]\!]$$

as weighted transition systems up to structural congruence, where u_1 is the first ancilla pattern of gadget i. Hence $\Pr_{[\![M]\!]}(y \mid \mathsf{do}(V_i = v)) = \Pr_M(y \mid \mathsf{do}(V_i = v))$ for every y.

Proof. At gadget i the restricted clause is the original clause times $\lceil *u_1 = v \rceil$, which is $P_i(v \mid pa)$ on the $(s-1)!$ candidates in the class $u_1 \mapsto v$ and 0 elsewhere. The contention set is the gadget alone (Lemma 5.4), so normalisation divides by $(s-1)!\,P_i(v \mid pa)$, which is nonzero by positivity, and the class fires with probability 1 *for every value of the parents*. The gadget's normalised dynamics is therefore the point mass at v independent of pa, which is exactly the mechanism of $M_{V_i=v}$ at V_i; the parents are still consumed, so the term is unchanged elsewhere; and every other gadget is unaffected by Proposition 4.4, the model term being flat. Theorem 5.5 applied to $M_{V_i=v}$ gives the distribution. □

The theorem is the reason to address the *ancilla* binding. Restricting a parent pattern p_j instead would select on inputs — a conditioning — not sever a mechanism. The embedding fixes that the value of a variable is carried by its ancilla binding, and that is the whole of what makes the surgery exact: the gadget's own gate is $\top$, so restriction and override coincide (Observation 4.6), and the weights that encoded the mechanism are the thing normalisation discards.

Remark 5.8 (Positivity is starvation). If $P_i(v \mid pa) = 0$ for some reachable pa, the restricted gadget has clause 0 on every candidate when those parents arrive, and the receipt waits forever: the history deadlocks by starvation, in the sense of [23]. This is exactly where Pearl's $\Pr(y \mid \mathsf{do}(x))$ is undefined without positivity, and the embedding renders the undefinedness as a mortal computation's death rather than as a division by zero. Positivity of the CPT is decidable by inspection on finite domains; the general question "is this do vacuous or fatal" is reachability (§7).

Corollary 5.9 (The do-calculus is derived)**.** *For positive M and disjoint $X, Y, Z, W \subseteq V$, the three rules of [1] hold of $[\![M]\!]$ under* do *as defined here:*

1. $\Pr(y \mid \mathsf{do}(x), z, w) = \Pr(y \mid \mathsf{do}(x), w)$ *if* $(Y \perp\!\!\!\perp Z \mid X, W)_{G_{\overline{X}}}$;
2. $\Pr(y \mid \mathsf{do}(x), \mathsf{do}(z), w) = \Pr(y \mid \mathsf{do}(x), z, w)$ *if* $(Y \perp\!\!\!\perp Z \mid X, W)_{G_{\overline{X}\underline{Z}}}$;
3. $\Pr(y \mid \mathsf{do}(x), \mathsf{do}(z), w) = \Pr(y \mid \mathsf{do}(x), w)$ *if* $(Y \perp\!\!\!\perp Z \mid X, W)_{G_{\overline{X}\,\overline{Z(W)}}}$,

with $Z(W)$ the Z-nodes that are not ancestors of W in $G_{\overline{X}}$, conditioning read as post-selection on histories, and do *on several variables as the conjunction & of their ancilla restrictions.*

Proof. By Theorems 5.5 and 5.7 and Proposition 4.8, every interventional distribution of $[\![M]\!]$ is the corresponding distribution of a submodel of M, and post-selection on histories is conditioning in that submodel. The rules are Pearl's theorem for the CBN M [1, 2]. □

The content of the corollary is not that the rules are true — Pearl proved that — but that the object they are stated for is literally present in the semantics, so that nothing is being modelled. The d-separation premises are statements about G, and G is the causal order of the staged term (Lemma 5.4); we conjecture in §10 that they are expressible as crisp-sort behavioural formulae over $[\![M]\!]$ and so dischargeable by the model checker.

5.4 Counterfactuals

Pearl's third rung asks what Y would have been had X been x, in the world that actually occurred. In an SCM this is abduction (infer U from the evidence), action (mutilate), prediction (compute).

Definition 5.10 (Exogenous record)**.** A *run* of $[\![M]\!]$ is a causal history together with, at each gadget, the candidate that fired. Its *exogenous record* u is the vector of fired ancilla classes. A run is *replayed under* $\mathsf{do}(C)$ from a pre-state S by executing $\mathsf{do}(C)$ in S with the resolver instructed, at each gadget whose contention set is unchanged from the recorded run, to fire the recorded class if it is still a candidate of nonzero restricted weight, and otherwise to sample.

Theorem 5.11 (Counterfactuals are replay)**.** *Let M be positive, let u be the exogenous record of a run with outcome $(v_1, \ldots, v_n)$, and let S be the initial state. Then replaying under $\mathsf{do}(u_1^{(i)} = x)$ from S yields, at every gadget other than i and its descendants, the recorded value, and at i and its descendants the values $Y_{V_i=x}(u)$ of the SCM whose exogenous variables are the ancilla draws. On the staged fragment the "otherwise sample" branch of Definition 5.10 is never taken.*

Proof. Lemma 5.4: every gadget's contention set is itself, and its candidate set depends only on which parent values arrived, every ancilla value being present. So the recorded class is always a candidate; it has nonzero restricted weight by positivity except at gadget i, where the restriction forces x. Non-descendants of V_i receive the same parent values as in the run and fire the recorded class; descendants receive the mutilated values and fire their recorded class, which is $f_j(pa'_j, u_j)$ for the SCM in which u_j selects the class — the standard reduction of a CBN to an SCM by inverse-CDF sampling, with the class index as U_j. □

The "otherwise sample" clause is where the general case lives: off the staged fragment an intervention can change what rests on a channel and thereby which candidates exist, and "the same draw" is no longer well defined. That is the dynamic-graph obstacle of §10 seen from the third rung.

5.5 The dictionary, completed

Table 1 collects the correspondence. The right-hand column is the subject of §8.

Pearl	**graded rho over $\mathbb{R}_{\geq 0}$**	**f1r3node**
variable V_i	gadget i; value = fired ancilla class	a COMM event
parents PA_i	data on the channels the pattern and clause read	the contention set, from the pre-state
mechanism $P_i(\cdot \mid pa)$	the clause, in DNF	the `condition` field, pure evaluator
exogenous U_i	the resolver's draw at the ancilla race	the executing validator's resolution
DAG G	causal order of the staged term	—
$\Pr(y \mid x)$	post-selection on histories, $1/\Pr(x)$ retries	filtering finalised traces
$\Pr(y \mid \mathsf{do}(x))$	$\mathsf{do}(u_1 = x)$ in $[\![M]\!]$	elaboration rewrite / membrane
σ-intervention	$\mathsf{do}(y \models \psi)$	same
$Y_x(u)$	replay under do with the record held	fork from pre-state, replay the trace
positivity	no restricted receipt starves	no deadlock in the trace
d-separation	behavioural formula over the staged term (conj.)	the Softwhere checker

Table 1: The dictionary.

6 Relation to the σ-calculus

Pearl's atomic do replaces a mechanism by a constant. Correa and Bareinboim [5] generalise to *σ-interventions*, which replace the mechanism of X by an arbitrary new one $P^*(x \mid pa^*)$ — possibly stochastic, possibly conditional on a different parent set — and give a calculus whose rules are stated on the graph G_σ in which X's incoming edges are those of the *new* mechanism. Atomic do is the case $pa^* = \emptyset$ and P^* a point mass; the soft and conditional interventions of [7] are intermediate cases.

The construct of §4 is natively a σ-intervention, and the three forms it takes have three induced graphs.

Proposition 6.1 (Induced graphs)**.** *Let gadget i of $[\![M]\!]$ be placed under $\mathsf{do}(u_1 \models \psi)$. Then the restricted gadget is a σ-intervention on V_i whose new mechanism is $P_i(v \mid pa) \cdot \psi[\![v, \cdot]\!]$ normalised, and whose parent set in G_σ is*

$$\mathrm{PA}_i^* \;=\; \mathrm{PA}_i \cup \mathrm{fn}(\psi) \setminus \{u_1\}$$

when ψ is not an indicator of a singleton, and $\mathrm{fn}(\psi)\setminus\{u_1\}$ when it is. Here $\mathrm{fn}(\psi)$ are the names of variables (readout channels of other gadgets, or data in scope) that ψ's crisp subformulae read.

Proof. Restriction multiplies, so the new mechanism depends on pa through P_i unless the restriction leaves a single class, in which case normalisation discards P_i (Theorem 5.7); and it depends on whatever ψ reads. The parent set of a mechanism is the set of variables it reads. □

Three readings follow. The *set-valued* $\mathsf{do}(u_1 \in S)$ with $|S| \geq 2$ keeps the original edges: it conditions the mechanism on S, which is a σ-intervention in which P^* is P_i restricted and renormalised, and Pearl's three rules do not apply to it verbatim because no edge is deleted. The *graded* $\mathsf{do}(u_1 \models \psi)$ with ψ reading only u_1 reweights the mechanism by a function of the value — a dose, in the vocabulary of [21]. The *conditional* $\mathsf{do}(u_1 \models \varphi)$ with φ reading other variables is a policy, a dynamic regime, and its parents are written in the formula. In every

case the graph on which the σ-calculus rules are to be evaluated is read off the syntax of the constraint, which is the same point as the first consequence of §1: the parent set is syntactic, now for the intervention as well as for the model.

Remark 6.2 (Override as σ). Replacing rather than multiplying — $\psi_r \mapsto \psi$ — is a σ-intervention with $\mathrm{PA}^* = \mathrm{fn}(\psi)$ outright, and is the construct one wants when the site's own gate must be severed. It is expressible as a second mode of the membrane and lowers the same way; we leave it out of the primitive syntax because on the embedded models it is unnecessary (Theorem 5.7), and off them its complexity is that of §7 with no compensating structure.

Remark 6.3 (Cross-site constraints are excluded). A constraint relating y_1 and y_2 bound at *different* receipts would be a correlation across sites: a σ-intervention with shared exogenous noise, which is outside both Pearl's and Correa–Bareinboim's calculi and is, operationally, a join of the two receipts. Definition 4.1 excludes it by requiring each φ_i to be a constraint on y_i alone; $\&$ is a product of independent restrictions, which is Pearl's joint do.

7 Complexity, and the fragments that tame it

The membrane semantics of Definition 4.2 is simple to state and expensive to run. The expense comes from three independent sources, and a fourth question — whether the do does anything at all — is undecidable in general. We name each, then propose the fragments.

7.1 Four sources of cost

The constraint. φ ranges over the full crisp sort, which contains the modality $\langle K \rangle$ and the fixed point νX. A modality-free spatial constraint of depth d costs $O(2^d)$ to evaluate against a datum (the companion forager charges $\kappa_{\mathrm{sense}} = 2^d$ for exactly this). A modal constraint requires successor exploration to the budget β, exponential in β. A fixed point requires model checking against the reachable configuration graph, which is PSPACE-hard on finite-state fragments and undecidable on rho at large. And the constraint is evaluated *per candidate*, since the clause is.

The scope. Under the membrane semantics the set of constrained receipts is not known statically: a receipt arriving by reflection must be checked for a constrained binder when it arrives, and the check is a name comparison under the hygiene convention, which is cheap, but the *tracking* of scope through reduction — which membrane a given receipt is under, when membranes nest and received code carries its own — is a run-time cost proportional to the depth of nesting, paid on every COMM.

The contention set. Restriction removes candidates and so changes the normalisation of the contention set they belonged to. Where contention sets are singletons (Lemma 5.4) this is a division. Where they are not, the normaliser is a sum over maximal matchings, which is permanent-shaped and #P-hard in general [17]; a do that touches a large contention set forces that sum to be recomputed.

Vacuity and fatality. Whether any receipt binding y ever fires inside P is reachability, undecidable in general. So whether a given do is vacuous (no constrained race ever arises), effective (some arise and some candidate always satisfies φ), or fatal (a constrained race arises with no satisfying candidate, and starves) cannot be decided for arbitrary P. This is the halting argument of §4 returning as a cost: the programmer cannot in general know what an intervention will cost before paying for it.

7.2 Fragments of the constraint language

Definition 7.1 (Constraint fragments). $\mathcal{F}_0$ **(values).** $y = Q$ and $y \in S$ for a finite set S of closed terms; and $\bigoplus_i v_i \otimes \lceil y = Q_i \rceil$. Evaluation is a structural-congruence test against $|S|$ terms: constant in the term, linear in the datum. This is the Pearl fragment: Theorems 5.7 and 5.11 and Corollary 5.9 use nothing else.

$\mathcal{F}_1$ **(spatial).** Modality-free, fixed-point-free crisp formulae over the bound datum and the data in scope, of bounded depth d; and graded formulae built from them by $\otimes$, $\oplus$ and constants. Evaluation is $O(2^d)$ per candidate and needs no successor exploration; the graded members are posynomials in their constants [18, Prop. 5.2], so a learner can differentiate them. Conditional σ-interventions (Proposition 6.1) live here.

$\mathcal{F}_2$ **(budgeted modal).** $\mathcal{F}_1$ closed under $\langle K \rangle_{\mathbb{V}}$ to a declared depth β. Evaluation explores successors to depth β and is charged the inference budget already present in the calculus [16, Rem. 6.5]. Interventions conditional on what a site *could* do next — "restrict to candidates after which φ is reachable in two steps" — live here.

$\mathcal{F}_3$ **(fixed points).** The full crisp sort. Not evaluable at the site in general; admissible in *specification* position, where the model checker discharges it offline on a finite-state fragment, and then lowered to $\mathcal{F}_1$ by replacing the fixed point with the finite disjunction it unfolds to on that fragment.

Principle 7.2 (Where each fragment runs). $\mathcal{F}_0$ and $\mathcal{F}_1$ run at the site, in the clause evaluator, at the cost of a guard. $\mathcal{F}_2$ runs at the site under the inference budget and is priced by it. $\mathcal{F}_3$ belongs to the checker, not the clause: it is what one *verifies* about an intervention, not what one *executes* as one. A `do` whose constraint is in $\mathcal{F}_3$ is a specification awaiting lowering, and Remark 4.9 says that lowering is the construct's whole job.

7.3 Type-based constraints on the scope

The constraint fragments bound the cost of one evaluation. The scope disciplines bound how many evaluations there are, whether the scope is static, and whether the contention sets are small. They are types in the sense the programme already uses: spatial-behavioural formulae imposed on P as admission conditions, decidable on the fragment and checked once.

Definition 7.3 (Scope disciplines). $\mathcal{T}_1$ **(staged).** Every channel in scope has exactly one receiver, every receipt consumes every message on each of its channels, and data flow uses the linear binding arrow $\circ\!\!-$ of [17] outside continuations' fan-out. Consequence: contention sets are singletons, the normaliser is a division, the causal order is a static DAG (Lemma 5.4), and each gadget fires once.

$\mathcal{T}_2$ **(flat).** No drop $*z$ in scope at a position that can come to contain a receipt; equivalently the stratified `where` of [20] with ceiling $\bot$ on binders. Consequence: the set of constrained receipts is static and lowering is exact (Proposition 4.4); no scope tracking at run time.

$\mathcal{T}_3$ **(finite-domain).** Every channel a constrained variable is bound on carries a DDL type with finitely many closed values. Consequence: $\mathcal{F}_0$ constraints are sets of enumerable size; vacuity and fatality are decidable by inspecting which values the clause assigns positive weight, which is the positivity check of Remark 5.8.

$\mathcal{T}_4$ **(confined).** Every constrained receipt lies within the deployer's seal in the sense of [24], so that the `do` reaches only code its author pays for. Consequence: the membrane coincides with the seal tree, the capability to intervene is the capability to deploy, and Proposition 3.3 is not a limitation but the boundary of the namespace.

constraint	per-candidate cost	needs	supports
$\mathcal{F}_0$	$O(\|S\|)$ congruence tests	nothing	Pearl's do, rules, counterfactuals
$\mathcal{F}_1$	$O(2^d)$	bounded depth d	conditional σ, doses, policies
$\mathcal{F}_2$	$O(b^\beta)$, b branching	inference budget β	interventions on reachable futures
$\mathcal{F}_3$	checker	finite-state fragment	specification only; lower to $\mathcal{F}_1$
scope type	**what it buys**		
$\mathcal{T}_1$ staged	singleton contention sets; static DAG; one firing per gadget		
$\mathcal{T}_2$ flat	exact lowering; no run-time scope tracking		
$\mathcal{T}_3$ finite-domain	decidable vacuity and fatality		
$\mathcal{T}_4$ confined	do reaches only paid-for code; the capability to intervene is the capability to deploy		

Table 2: Fragments and disciplines.

Proposition 7.4 (The practical fragment)**.** *Under $\mathcal{T}_1$–$\mathcal{T}_4$ with constraints in $\mathcal{F}_0$: (i) $\mathsf{do}(C)$ in P lowers at elaboration and the membrane is erased; (ii) each constrained candidate costs one structural-congruence test against $|S|$ closed terms; (iii) vacuity and fatality are decidable; (iv) the induced graph G_σ is the static DAG with the constrained variables' incoming edges deleted; and (v) the semantics is that of Definition 5.1 exactly. Under $\mathcal{T}_1$–$\mathcal{T}_3$ with constraints in $\mathcal{F}_1$, (i)–(iii) hold, G_σ is the static DAG with PA_i^* as in Proposition 6.1, and the semantics is a σ-intervention on a CBN.*

Proof. (i) is Proposition 4.4, whose hypothesis is $\mathcal{T}_2$. (ii) is the definition of $\mathcal{F}_0$. (iii): under $\mathcal{T}_1$ every constrained receipt fires exactly once and its candidate set is determined by its parents' values, which range over a finite product by $\mathcal{T}_3$; enumerate. (iv) and (v) are Theorem 5.7 and Proposition 6.1, whose hypotheses are supplied by $\mathcal{T}_1$ and $\mathcal{T}_3$. The second statement replaces Theorem 5.7 by Proposition 6.1. □

The disciplines are not new to the programme. $\mathcal{T}_1$ is the safe fragment of [17, §9] in which $[\![P]\!]$ is a quantum state; $\mathcal{T}_2$ is the flat fragment of [21] and the stratified one of [20]; $\mathcal{T}_3$ is what the DDL supplies for free; and $\mathcal{T}_4$ is the seal discipline of the cost-accounting work. That four notes written for four purposes draw the same line is the best evidence we have that the line is real. The route to larger fragments is the one the monadic-foreach note already proposes: typed calculi translated into the language with types as spatial-behavioural formulae, System F first and then Abramsky's CLL, each of which supplies a terminating fragment on which $\mathcal{F}_3$ is decidable.

Remark 7.5 (What is lost outside the fragment). Outside $\mathcal{T}_1$ the contention sets are the branching cells of [10] and a do that touches one must be understood as a restriction of the cell's distribution; outside $\mathcal{T}_2$ a do is a genuine run-time membrane; outside $\mathcal{T}_3$ positivity is a reachability question; outside $\mathcal{T}_4$ an intervention on another's code is a soft intervention by Proposition 3.3, an experiment rather than a surgery. None of these is a failure of the construct. They are where Pearl's calculus stops and the calculus of this programme continues.

8 Execution

The construct was designed against the execution framework rather than for a simulator, and most of what it needs exists.

Lowering. The normaliser already carries an elaboration pass that puts every deployment into normative forms so that the reducer's seal record is read off the term. Lowering do under

$\mathcal{T}_2$ is one more rewrite in that pass, and it leaves no run-time trace: the constrained clauses are ordinary clauses with one more conjunct, evaluated by the pure `condition` evaluator the matcher already calls per candidate.

The membrane. Off $\mathcal{T}_2$ the `do` is a run-time membrane, and it is the twin of the cost membrane $\{\%\ P\ \%\}[s]$: a scoped decoration on the cut, read off the term by the reducer, altering the COMM rule within. The seal tree rooted at the deployer's wallet becomes a seal-and-constraint tree; a receipt arriving by reflection inherits the innermost constraint as it inherits the innermost seal. The two decorations commute (Proposition 4.8 extends to them, since metering does not read the clause and the clause does not read the meter) and so may be nested in either order.

Pricing. An intervention is one COMM on the port, by Principle 3.4, plus the marginal cost of the extra conjunct per candidate evaluation, which under $\mathcal{F}_0$ is a congruence test. The charge for an $\mathcal{F}_2$ constraint is the inference budget it declares. Fixing these against the ledger is Obligation 10.3.

Observation, and who does it. The validators' execution model — one node executes and produces a trace, the others replay it to confirm the state change — is, in the vocabulary of [22], an experimenter/spectator split. The executing validator resolves the races and so performs rung two; the replayers observe and are rung one. $\Pr(y \mid x)$ is estimated from the finalised trace; it costs nothing beyond storage and is exactly what [22, Prop. 4.1] says it is: the trace distribution, and no more.

Counterfactuals. Theorem 5.11 needs a pre-state and a record of fired candidates. The order-then-execute design runs each envelope from the block pre-state and records the winners; the losing candidates are recomputable from the pre-state deterministically, so nothing further need be stored. The same record is what the gradient of [18, Cor. 5.7] accumulates — one number per firing from the candidates the site already evaluated — so one instrumentation serves learning and counterfactual replay. The stable site identity the replay requires is the quotation of the receipt, exactly as the heralds of [16, §9] name their sites.

The kernel. $\Pr(y \mid \mathsf{do}(x))$ is $e_y^\top T^{\beta}_{\mathsf{do}(x)} |P\rangle$ for the transfer operator with one column block replaced, and the adjoint of [18, Thm. 5.6] is the same Neumann series. A batch of interventional queries and the modal gradient share one sparse matrix-vector path on the GPU.

9 Inhabiting choice, discharged

The companion note [22] makes one observation three times: a system whose witnesses reduce confluently can *state* a non-confluent phenomenon but cannot *inhabit* it, so its identity on witnesses is forced to be linear-time, and the logic adequate for that identity is forced to be determinate. In its reading for artificial intelligence it aligns the linear/branching/causal spectrum with Pearl's ladder — association, intervention, counterfactual — and names the remedy as witnesses that can be races and an identity on them at least as fine as step bisimulation. The alignment is by argument. The present note is its constructive content, and we propose the mechanism as the tool for each of the obligations that note leaves owed.

Rung two is a graded race under a do. The note's rung two is "what an experimenter supplying contexts can learn," characterised by context-labelled bisimulation. Proposition 3.3 sharpens this: a context supplies apertures and cannot force them, so the experimenter's power

over $\mathbb{R}_{\geq 0}$ is exactly the do membrane on the code it owns together with soft interventions on the rest. Theorem 5.7 says that on the staged fragment that power is Pearl's rung two on the nose. Rung three is Theorem 5.11: replay with the record held.

The vending-machine benchmark becomes a program. The smallest behaviour invisible to a trace is Milner's pair

$$P = coin.(tea + coffee), \qquad Q = coin.tea + coin.coffee,$$

and the observable a trace cannot recover is a *refusal*: coin in, tea requested, nothing out. Render the machines with races in the usual way, with y the variable bound at the drink-selecting receipt. Then

$$\mathsf{do}(y = tea)\ \mathsf{in}\ (coin!() \mid P) \quad \text{and} \quad \mathsf{do}(y = tea)\ \mathsf{in}\ (coin!() \mid Q)$$

differ in exactly one way: the first always delivers tea, and the second, on the branch where Q has committed to coffee before the coin, has a restricted receipt with no satisfying candidate and *starves*. The refusal that never appears in a log of completed runs is a deadlock in the intervened term. The benchmark the note proposes — logs from one machine chosen by a hidden coin, completed actions only, then the refusal question — is answered by one do and one observation of whether anything was delivered, which is the "one attempted tea after one coin" the note says settles the question; here it is a term one can run.

The audit as a theorem. The note's verification reading says a λ-term prover can certify a concurrent system only relative to a stipulated model of its non-determinism, and that the faithfulness of the model is not auditable by any means the prover possesses. Its obligation is to say which class of tests such a prover can run and prove the class contained in trace observation. Proposition 3.3 is half of that statement: a test that cannot force a race is a context, and a context's observations are contained in the support reading. The other half is that a do is precisely a test outside that class, and that on a process runtime the certificate can contain one.

The three requirements as a theorem. The note's closing section argues that recursive self-improvement needs an improvement operator that acts on witnesses that are races, preserves branching through evaluation, and funds the branches that lose. A do is an operator on witnesses that changes their branching — it is the only kind there is, by §3 — and the set-valued form $\mathsf{do}(y \in S)$ with S the *losing* candidates is the operator that funds them: it restricts the race away from its current winner. That an improvement operator definable on a determinate language cannot increase branching is Proposition 3.2 read for the whole language; that one definable on rho with the cost endofunctor can is this note.

Hedges that are inherited. The note's last row asks what leaves the saddle point: tactics as witnesses, certificates that run, hedges that are inherited. A hypothesis of the mortal scientist is a clause; a clause under a do is a hypothesis together with the experiment that would test it; and a germ carrying both is a hedge that is inherited. Which is to say that the construct of this note is what the scientist's germs were missing in order to be rung-two objects rather than rung-one ones, and the reproduction note's crossover on quoted code applies to it unchanged, since a do is a term.

10 Open problems

Obligation 10.1 (d-separation as a behavioural formula)**.** Corollary 5.9 evaluates the d-separation premises on G. Since G is the causal order of the staged term, the premises should be crisp-sort formulae over $[\![M]\!]$ — "no history has a causal path from a Z-firing to a Y-firing not passing through X or W" is a ν-property — dischargeable by the model checker. Stating and proving the correspondence would make the rules' side conditions checkable in the same instrument as their conclusions.

Obligation 10.2 (The rules off the staged fragment)**.** Outside $\mathcal{T}_1$ the contention sets are branching cells and the event structure may be confused. The conjecture is that the σ-calculus rules hold on the confusion-free fragment with d-separation read off the event structure, and that on the confused fragment they hold per branching cell with the cell's distribution in place of the mechanism. The results of [9, 10] are the starting point and should be verified against the sources before being leaned on.

Obligation 10.3 (The price of a do)**.** Principle 3.4 says an intervention is a COMM. The marginal charge of the constraint conjunct, the budget charge of an $\mathcal{F}_2$ constraint, and the charge for a counterfactual replay have not been fixed against the ledger. Until they are, "an intervention costs one phlo" names an intention.

Obligation 10.4 (Identifiability across namespaces)**.** Pearl's identifiability question — when is $\Pr(y \mid \mathsf{do}(x))$ computable from observational data in the presence of hidden variables — has a sharp rendering here: the hidden variables are the code one cannot read, the boundary is the namespace, and the observational data is the shard's finalised trace. The ID algorithm of [6] should transfer to the staged fragment directly; what it says about a shard is the question.

Obligation 10.5 (Continuous domains)**.** The gadget enumerates the domain on the ancilla channel. A continuous variable needs the DDL's numeric types to carry an ancilla as a density, and the contention-set normalisation to become an integral. The Gillespie reading of [25] is the natural home.

Obligation 10.6 (Cycles)**.** A recursive term has an acyclic event-level graph and a cyclic variable-level one. The cyclic SCMs of [8] and the graded fixed point of [18] should meet here, with the budget β as the unrolling depth.

11 Conclusion

A graded program over the non-negative reals was a causal model before anyone said so: its likelihood factorises over sites, its parents are written in its clauses, and its exogenous noise is the resolver the programme had already made a parameter. What it lacked was an operator that forces a race, and the operator turns out to want a specific home. It cannot sit at a node, because whether a node will ever occur is undecidable; it sits at the cut, as a policy conditional on a race arising, and it lowers into the where-clause by the normal-form theorem for weight maps, so it adds nothing to the calculus that was not already expressible — only a place to say it.

With it, Pearl's calculus embeds exactly, and the embedding is the Shor gadget with a stochastic matrix in place of a unitary one. Intervening on the ancilla binding deletes incoming edges because normalisation discards the weights that encoded the mechanism; positivity is a receipt that does not starve; the three rules are theorems; counterfactuals are replay. Set-valued and graded constraints land in the σ-calculus with the induced graph read off the free names of the constraint. The construct is expensive to run naively, in ways that are each

separately familiar from the programme's other notes, and the fragments that tame it are the fragments those notes had already drawn.

The mechanism is offered as the constructive half of the inhabiting-choice argument. A race under a `do` is rung two; its replay is rung three; and the refusal that no corpus of completed runs can reveal is, in the intervened term, a receipt that waits for a tea that will never come.

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